

MALEXANDER, Sweden

WMO: 025520

Lat: 58.072N

Lon: 15.233E

Elev: 646

StdP: 14.36

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.4	10.5	1.9	6.2	7.3	6.5	7.8	13.0	20.0	41.6	17.8	38.5	2.7	20	0.572

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.2	81.2	63.9	77.7	62.8	74.0	61.5	66.8	75.6	64.9	73.2	63.2	71.0	6.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.9	91.0	69.2	62.0	85.1	67.5	60.2	79.8	66.1	31.7	75.0	30.3	73.3	29.0	70.8	73.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.8	13.9	12.2	DB	1.0	85.3	7.2	3.7	-4.1	87.9	-8.3	90.0	-12.3	92.1	-17.5	94.8	(4)
(5)			WB	0.2	69.7	6.8	2.3	-4.7	71.4	-8.7	72.7	-12.6	74.0	-17.6	75.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.4	27.8	28.0	33.5	42.3	50.9	58.3	63.2	61.0	53.3	44.4	37.0	31.7	(6)	
	DBStd	14.06	8.28	8.19	7.75	6.43	6.07	5.46	5.47	4.82	5.41	6.67	6.74	8.23	(7)	
	HDD50	3307	689	617	512	243	60	3	0	1	28	195	390	569	(8)	
	HDD65	7608	1154	1037	977	681	438	213	104	142	350	639	839	1034	(9)	
	CDD50	1256	0	0	0	12	89	252	409	343	129	21	1	0	(10)	
	CDD65	80	0	0	0	0	1	12	47	19	1	0	0	0	(11)	
	CDH74	754	0	0	0	0	18	157	444	131	4	0	0	0	(12)	
	CDH80	142	0	0	0	0	0	23	98	21	0	0	0	0	(13)	
(14) Wind	WSAvg	5.9	6.4	6.3	6.4	5.9	5.7	5.6	5.1	5.1	5.6	5.9	6.6	6.5	(14)	
(15) Precipitation	PrecAvg	23.1	1.4	1.3	1.1	1.3	1.8	2.6	3.1	2.8	2.1	2.0	2.0	1.6	(15)	
	PrecMax	28.9	2.7	2.1	2.0	3.1	3.2	6.0	5.6	6.1	5.0	5.0	3.6	2.9	(16)	
	PrecMin	17.7	0.4	0.2	0.2	0.1	0.3	0.0	0.4	0.7	0.5	0.6	0.6	0.3	(17)	
	PrecStd	2.9	0.7	0.4	0.4	0.9	0.8	1.4	1.5	1.4	1.3	1.0	0.8	0.7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.8	48.4	57.8	67.6	76.5	83.1	87.0	82.1	73.3	62.7	52.9	50.2	(19)	
		MCWB	42.7	43.3	46.3	52.4	59.9	62.5	65.6	65.5	62.5	55.8	50.8	47.8	(20)	
	2%	DB	42.1	43.3	52.1	63.0	71.0	78.8	82.9	77.6	68.7	58.2	50.1	46.8	(21)	
		MCWB	40.1	40.0	43.5	50.1	56.9	62.0	64.9	63.8	60.0	54.0	48.2	44.7	(22)	
	5%	DB	40.0	41.1	47.3	58.5	66.9	74.4	79.3	73.6	64.8	55.6	48.1	44.2	(23)	
		MCWB	38.3	38.5	41.2	47.8	54.5	60.1	63.6	62.9	57.8	52.4	46.3	42.3	(24)	
	10%	DB	37.9	38.9	43.6	54.3	62.9	70.4	75.6	70.3	61.7	53.4	46.1	41.6	(25)	
		MCWB	36.4	36.7	39.3	45.6	52.4	57.9	62.6	61.1	56.2	50.5	44.5	39.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.3	44.1	47.6	54.9	62.8	66.6	69.5	68.6	64.5	56.8	51.4	48.3	(27)	
		MCDB	44.5	46.6	54.4	66.1	72.5	75.2	79.5	76.3	70.1	59.7	52.3	49.8	(28)	
	2%	WB	40.5	40.8	44.4	51.7	59.3	63.5	67.4	66.2	61.0	54.7	48.4	44.8	(29)	
		MCDB	41.7	43.1	50.7	59.6	68.0	74.5	77.2	73.2	66.3	57.6	49.7	46.3	(30)	
	5%	WB	38.5	38.7	42.0	49.0	56.5	61.1	65.7	64.1	58.8	52.7	46.6	42.4	(31)	
		MCDB	39.8	40.8	46.3	56.6	63.5	72.0	75.1	70.4	63.7	55.3	47.9	43.9	(32)	
	10%	WB	36.5	36.7	39.8	46.6	53.8	59.2	63.9	62.5	56.8	50.9	44.6	40.0	(33)	
		MCDB	37.8	38.7	43.2	53.3	60.7	68.6	72.6	68.5	60.9	53.1	46.0	41.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.6	9.1	12.7	17.5	18.0	17.7	17.2	15.1	13.4	10.2	7.1	6.9	(35)	
		MCDBR	8.6	10.0	18.5	24.3	24.4	25.1	24.5	20.3	17.6	11.7	8.4	8.7	(36)	
	5% WB	MCWBR	8.0	7.8	11.6	13.8	12.7	12.1	9.8	9.6	10.0	8.2	7.7	8.3	(37)	
		MCWBR	8.4	8.9	16.9	20.7	20.5	21.8	20.4	17.3	16.2	11.0	8.1	8.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.277	0.312	0.331	0.369	0.366	0.355	0.379	0.372	0.342	0.333	0.303	0.263	(40)	
		taud	2.254	2.313	2.372	2.338	2.408	2.460	2.409	2.426	2.497	2.442	2.318	2.189	(41)	
	Ebn at Noon		183	229	259	266	276	280	269	263	255	221	170	146	(42)	
		Edh at Noon	15	22	28	34	33	32	33	30	24	20	14	11	(43)	
(44) All-Sky Solar Radiation	RadAvg		113	311	787	1262	1630	1794	1611	1290	876	401	146	79	(44)	
	RadStd		18	55	88	154	149	88	196	135	121	68	27	8	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (54 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	-230	-281	N/A	N/A

Nomenclature: See separate page